Bose-Einstein condensation of magnons

$$H_{AFM} = \frac{1}{2} J \sum_{r,g} \vec{S}_r \cdot \vec{S}_{r+g} - \sum_r h S_r^z$$

For sufficiently large $h > h_c$ (to be determined below) all spins point out in the direction of $\vec{h}$ $\Rightarrow$ fully polarized state $|F\rangle = |\uparrow\uparrow\uparrow\ldots\rangle$ is the <u>ground state</u>.

$$S_r^z = S - a_r^\dagger a_r$$

$$S_r^+ = \sqrt{2S}\sqrt{1 - \frac{a_r^\dagger a_r}{2S}}\, a_r \cong \sqrt{2S}\left(a_r - \frac{1}{4S} a_r^\dagger a_r a_r\right)$$

$$H_{AFM} = \frac{1}{2} J \sum_{r,g} \Big\{ (S - a_r^\dagger a_r)(S - a_{r+g}^\dagger a_{r+g})$$

$$+ \frac{1}{2}(2S)\Big[\left(a_r - \frac{1}{4S} a_r^\dagger a_r a_r\right)\left(a_{r+g}^\dagger - \frac{1}{4S} a_{r+g}^\dagger a_{r+g}^\dagger a_{r+g}\right)$$

$$+ h.c.\Big]\Big\} - h \sum_r (S - a_r^\dagger a_r) = H_{AFM}^{(0)} + H_{AFM}^{(2)} + H_{AFM}^{(4)}$$

$$H^{(2)} = \frac{1}{2} J \sum_{r,g} (-S)\left[a_r^\dagger a_r + a_{r+g}^\dagger a_{r+g} - a_r a_{r+g}^\dagger - a_{r+g}^\dagger a_r\right]$$

$$= -\frac{1}{2} J S \sum_g \sum_k \left\{ 2 a_k^\dagger a_k - \left(e^{i\bar{k}\bar{g}} + e^{-i\bar{k}\bar{g}}\right) a_k^\dagger a_k \right\}$$

$$= -JS \sum_k \underbrace{\sum_g \left(1 - \cos(\bar{k}\cdot\bar{g})\right)}\, a_k^\dagger a_k$$

(also need to add mag. field term)

for $d=3$ cubic lattice:

$$2d - 2(\cos k_x a + \cos k_y a + \cos k_z a)$$

$$= 6\left[1 - \frac{1}{3}(\cos k_x a + \cos k_y a + \cos k_z a)\right]$$

Full harmonic Hamiltonian also includes term $h \sum_r a_r^+ a_r = h \sum_k a_k^+ a_k$.

$$\Rightarrow H^{(2)} = \sum_k \underbrace{\left[h - 6JS\left(1 - \frac{1}{3}\sum_{j=1}^{3} \cos(k_j a)\right)\right]}_{\mathcal{E}(k)} a_k^+ a_k .$$

$\mathcal{E}(k)$ = this is dispersion of spin-S antiferromagnet is a fully polarized state!

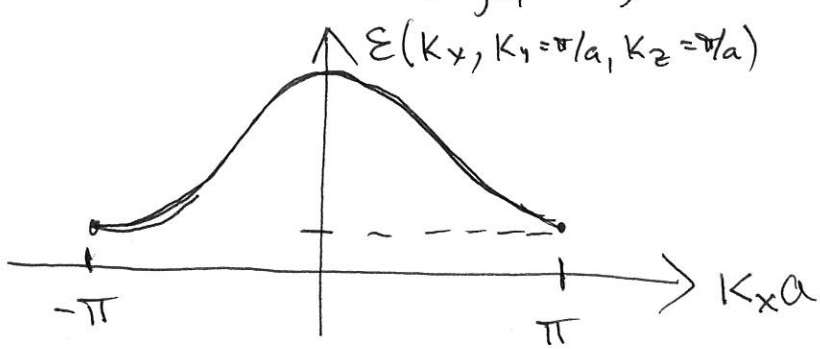
Find minimum: $\frac{\partial \mathcal{E}(\vec{k})}{\partial k_j} = 0 \Rightarrow \begin{cases} 2JS(-\sin k_x a) = 0 \\ -\sin k_y a = 0 \\ -\sin k_z a = 0 \end{cases}$

But, for the minimum, we also need $\frac{\partial^2 \mathcal{E}}{\partial k_j^2} > 0$

$$\Rightarrow \begin{matrix} -\cos k_x a > 0 \\ -\cos k_y a > 0 \\ -\cos k_z a > 0 \end{matrix} \Rightarrow \begin{cases} k_x = \frac{\pi}{a} \\ k_y = \pi/a \\ k_z = \pi/a \end{cases}$$

At the minimum $\vec{k}_0 = \frac{\pi}{a}(1,1,1)$ and ~~$\mathcal{E}(\vec{k}_0) =$~~

$$\mathcal{E}(\vec{k}_0) = h - 6JS\left(1 - \frac{1}{3}\sum_{j=1}^{3}(-1)\right) = h - 12JS$$



<u>Next</u>: compare this with standard BEC

Expand near the minimum (where particles are at $T \to 0$)

$$\varepsilon(\vec{k}_0 + \vec{q}) \simeq \varepsilon(\vec{k}_0) + \vec{q} \cdot \vec{\nabla} \varepsilon(\vec{k}_0) + \frac{1}{2} q_j^2 \partial_j^2 \varepsilon(\vec{k}_0) + \ldots$$

$$= h - 12JS - 2JSa^2 \sum_{j=1}^{3} \frac{1}{2} q_j^2 \underbrace{\cos(k_j a)}_{-1}$$

$$= h - 12JS + JSa^2 (q_x^2 + q_y^2 + q_z^2) + \ldots$$

$$\Rightarrow H^{(2)} \approx \sum_q \left(h - 12JS + JSa^2 \vec{q}^{\,2} \right) a^+_{k_0+q} a_{k_0+q}$$

Note $a_r = \frac{1}{\sqrt{N}} \sum_k a_k e^{i\vec{k}\vec{r}} \approx \frac{1}{\sqrt{N}} \sum_q a_{k_0+q} e^{i\vec{k}_0 \cdot \vec{r} + i\vec{q} \cdot \vec{r}}$

$$= e^{i\vec{k}_0 \cdot \vec{r}} \frac{1}{\sqrt{N}} \sum_q a_{k_0+q} e^{i\vec{q} \cdot \vec{r}}$$

Can denote $a_{k_0+q} = b_q$.

$$H^{(2)} \simeq \sum_q (\varepsilon_q - \mu) a^+_{k_0+q} a_{k_0+q}$$

$$= \sum_q (\varepsilon_q - \mu) b_q^+ b_q$$

where $\mu = \underbrace{12JS}_{\text{Saturation field}} - h = h_{sat} - h$

Note $\mu < 0$ for $h > h_{sat}$, but $\mu > 0$ for $h < h_{sat}$!

Recall standard BEC : bosons with $\varepsilon_p = \frac{\vec{p}^2}{2m}$

$\Rightarrow$ occupation number of state $|\vec{p}\rangle$ is

$$f(\vec{p}) = \frac{1}{e^{\frac{\varepsilon_p - \mu}{k_B T}} - 1}$$, where μ = chemical potential.

Hence $f(\vec{p}=0) = \frac{1}{e^{-\frac{\mu}{k_B T}} - 1}$ which implies

that $-\mu \geq 0$ always (as otherwise can get $f(\vec{p}=0) < 0$ at low T)

$$\Rightarrow \boxed{\mu \leq 0}$$

As we keep lowering T, at some T_0

$\mu(T_0) = 0$ at which point occupancy of $|\vec{p}=0\rangle$

state becomes ~~inter.~~ <u>macroscopic</u>

fixed $$N = \sum_{\vec{p}} f(\vec{p}) = N_0(T) + \int \frac{d^3\vec{p}}{(2\pi\hbar)^3} f(\vec{p})$$

T-dependent

$$\frac{N_0(T < T_0)}{N} \sim O(1)$$

This is Bose-Einstein condensation (BEC)

Let us work out interaction term now:

$$\mathcal{H}^{4z} = \frac{1}{2} J \sum_{rg} a_r^+ a_r \, a_{r+g}^+ a_{r+g} =$$

from $S^z S^z$

$$= \frac{1}{2} J \sum_{rg} \frac{1}{N^2} \sum_{\substack{k_1 k_2 \\ k_3 k_4}} e^{i\vec{r}(\vec{k}_1+\vec{k}_2)} a_{k_1}^+ a_{k_2} \, e^{i(\vec{r}+\vec{g})(-\vec{k}_3+\vec{k}_4)} a_{k_3}^+ a_{k_4}$$

$$= \frac{1}{2} J \sum_g \underbrace{\sum_r e^{i\vec{r}(\vec{k}_1+\vec{k}_3-\vec{k}_2-\vec{k}_4)}}_{N\,\delta_{\vec{k}_1+\vec{k}_3,\,\vec{k}_2+\vec{k}_4}} \frac{1}{N^2} \sum_{(\vec{k}_i)} e^{-i\vec{g}(\vec{k}_3-\vec{k}_4)} a_{k_1}^+ a_{k_3}^+ a_{k_2} a_{k_4}$$

let $\vec{k}_4 = \vec{k}'$, $\vec{k}_2 = \vec{k}$ $\Rightarrow$ $\vec{k}_1 = \vec{k}+\vec{q}$, $\vec{k}_3 = \vec{k}'-\vec{q}$

$$\mathcal{H}^{4z} = \frac{1}{2} J \frac{1}{N} \sum_{\vec{k},\vec{k}',\vec{q}} \sum_g e^{i\vec{q}\cdot\vec{g}} a_{k+q}^+ a_{k'-q}^+ a_{k'} a_k$$

$$= \frac{1}{2N} \sum_{\substack{\vec{k},\vec{k}' \\ \vec{q}}} V_z(\vec{q}) \, a_{k+q}^+ a_{k'-q}^+ a_{k'} a_k$$

$$V_z(\vec{q}) = J \sum_{\vec{g}} e^{i\vec{g}\cdot\vec{q}} = 2J(\cos q_x a + \cos q_y a + \cos q_z a)$$

Let's denote it as $J(q) = J \sum_{j=1}^{3} \cos(q_j a)$

keep only quartic part here.

$$H^{4\perp} = \frac{1}{2} J \sum_{rg} \frac{1}{2}\left(\sqrt{2S}\right)^2 \Big\{ \left(a_r - \frac{1}{4S} a_r^+ a_r a_r\right) \cdot$$

from $S^x \cdot S^x + S^y \cdot S^y$

$$\cdot \left(a_{r+g}^+ - \frac{1}{4S} a_{r+g}^+ a_{r+g}^+ a_{r+g}\right) + h.c. \Big\}$$

$$= \frac{-1}{4} \cdot \frac{1}{2} J \sum_{rg} \left(a_r^+ a_r a_r a_{r+g}^+ + a_r a_{r+g}^+ a_{r+g}^+ a_{r+g} + h.c. \right)$$

$$= -\frac{1}{8} J \sum_{rg} \frac{1}{N^2} \sum_{\vec{k}_i} \Big(e^{i\vec{r}(-\bar{k}_1 + \bar{k}_2 + \bar{k}_3 - \bar{k}_4)} \underbrace{a_{k_1}^+ a_{k_2} a_{k_3} a_{k_4}}_{e^{-i\vec{g}\cdot\vec{k}_4}}$$

$$+ e^{i\vec{r}(\bar{k}_1 - \bar{k}_2 - \bar{k}_3 + \bar{k}_4)} \underbrace{a_{k_1} a_{k_2}^+ a_{k_3}^+ a_{k_4}}_{e^{i\vec{g}\cdot\vec{k}_4}} + h.c. \Big)$$

$$\left(\sum_r e^{\pm i\vec{r}(\bar{k}_1 + \bar{k}_4 - \bar{k}_2 - \bar{k}_3)} = N \delta_{k_1 + k_4, k_2 + k_3} \right)$$

$$= -\frac{1}{8} J \sum_{\vec{g}} \frac{1}{N} \sum_{\bar{k}, \bar{k}', \bar{q}} \Big(a_{k+q}^+ a_{k'-q}^+ a_{k'} a_k e^{-i\vec{g}\cdot(\vec{k}' - \vec{q})}$$

$$+ a_{k+q}^+ a_{k'-q}^+ a_{k'} a_k e^{i\vec{g}\cdot\vec{k}} + h.c. \Big)$$

$$= -\frac{1}{4} \frac{1}{2N} \sum_{\bar{k}, \bar{k}', \bar{q}} \Big(\left[J(\vec{k}) + J(\vec{k}' - \vec{q}) \right] a_{k+q}^+ a_{k'-q}^+ a_{k'} a_k + h.c. \Big)$$

It is easy to check that

<u>h.c.</u> terms give exactly the same contribution.

$$H^{(4)} = \frac{1}{2N} \sum_{\vec{k},\vec{k}',\vec{q}} \left[J(\vec{q}) - \frac{1}{2}\left(J(\vec{k}) + J(\vec{k}'-\vec{q}) \right) \right] \cdot$$

$$\cdot\, a^{+}_{k+q}\, a^{+}_{k'-q}\, a_{k'}\, a_{k}$$

$$H^{(2)} = \sum_{q} \left(\varepsilon_q - \mu \right) a^{+}_{k_0+q}\, a_{k_0+q}$$

Lowest energy state corresponds to $\vec{q}=0$ $(\varepsilon_{\vec{q}=0}=0)$ and for $\mu>0$ (which means <u>$h<h_{sat}$</u> which is easy to do in the lab) state with $\vec{q}=0$ must be <u>macroscopically occupied</u>

So we write $a_{k_0+q} = \underbrace{\sqrt{N}\,\psi}_{\text{condensate}}\, \delta_{\vec{q},0} + \underbrace{\tilde{a}_{k_0+q}}_{\text{here } \vec{q}=0 \text{ is excluded}}$

OR, $a_k = \sqrt{N}\,\psi\, \delta_{\vec{k},\vec{k}_0} + \tilde{a}_k$

Plug this into $H = H^{(2)} + H^{(4)}$

$$\boxed{\psi = \text{complex number}}$$

Condensate contribution

$$n_k = \langle a_k^+ a_k \rangle = \underbrace{N \psi^* \psi}_{N_0} \delta_{k,k_0} + \langle \tilde{a}_k^+ \tilde{a}_k \rangle$$

$$\frac{N_0}{N} = |\psi|^2 \Rightarrow \psi \text{ describes condensate fraction.}$$

$$\mathcal{H}^{(2)} \rightarrow \mathcal{H}_0^{(2)} \approx -\mu \sum_{\vec{k}=\vec{k}_0} \left(\sqrt{N}\,\psi^*\right)\left(\sqrt{N}\,\psi\right) = -\mu N |\psi|^2.$$

(Single term!)

$$\mathcal{H}^{(4)} \rightarrow \mathcal{H}_0^{(4)} = \frac{1}{2N} \sum_{\begin{pmatrix} \vec{k}=\vec{k}_0 \\ \vec{k}'=\vec{k}_0 \\ \vec{q}=0 \end{pmatrix}} \left[J(0) - \frac{1}{2}\left(J(\vec{k}_0) + J(\vec{k}_0)\right)\right] \cdot$$

$$\cdot \left(\sqrt{N}\,\psi^*\right)\left(\sqrt{N}\,\psi^*\right)$$

$$\cdot \left(\sqrt{N}\,\psi\right)\left(\sqrt{N}\,\psi\right)$$

$$= \frac{1}{2N}\left(\sqrt{N}\right)^4 \left(J(\vec{0}) - J(\vec{k}_0)\right) |\psi|^4$$

$$J(0) = 2J \cdot 3$$

$$J(\vec{k}_0) = 2J\left(\cos\pi + \cos\pi + \cos\pi\right) = -6J$$

$$\Rightarrow \mathcal{H}_0^{(4)} = N(6J)\,|\psi|^4$$

Thus the condensate energy is

$$E_0 = \mathcal{H}_0^{(2)} + \mathcal{H}_0^{(4)} = N\left\{-\mu|\psi|^2 + \tfrac{1}{2}u|\psi|^4\right\}$$

(where $u = 12J$ for generality of notations)

let $\psi = \sqrt{\rho}\, e^{i\varphi}$ where ρ = condensate density

$$\Rightarrow \frac{E_0}{N} = -\mu\rho + \tfrac{1}{2}u\rho^2$$

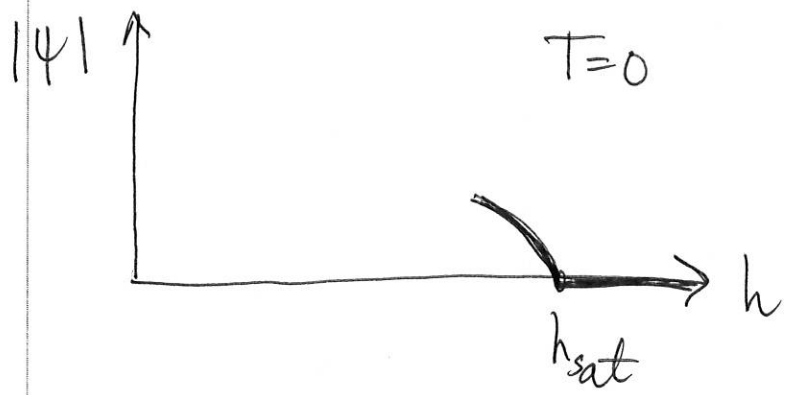
Optimal ρ_0 is found by minimizing E_0

$$\frac{\partial E_0}{\partial \rho} = 0 = -\mu + u\rho_0 \Rightarrow \rho_0 = \frac{\mu}{u}$$

$$\psi = \sqrt{\frac{\mu}{u}}\, e^{i\varphi} = \sqrt{\frac{h_{sat} - h}{u}}\, e^{i\varphi}$$

Condensate fraction is controlled by magnetic field!

Thus, ψ = ORDER PARAMETER



2nd order phase transition.

$$\frac{E}{N} = -\mu|\psi|^2 + \frac{1}{2}u|\psi|^4$$

Landau-Ginzburg (LG) free energy (if generalized to $T \gg 0$)

Describes order parameter.

Non-condensate modes $\tilde{a}_k$ with $\vec{k} \neq \vec{k}_0$ represent quantum fluctuations above condensate (to be studied later).

The meaning of BEC

$$S^z_r = S - a^+_r a_r \quad ; \quad S^+ \approx \sqrt{2S}\, a_r$$

$$a_r = \frac{1}{\sqrt{N}} \sum_k e^{i\bar{k}\cdot\bar{r}} a_k = \frac{1}{\sqrt{N}} \sum_k e^{i\bar{k}\cdot\bar{r}} \left(\sqrt{N}\,\psi\,\delta_{k,k_0} + \tilde{a}_k\right)$$

$$= \underbrace{\psi\, e^{i\vec{k}_0\cdot\vec{r}}}_{\text{condensate}} + \underbrace{\frac{1}{\sqrt{N}} \sum_k e^{+i\bar{k}\cdot\bar{r}} \tilde{a}_k}_{\text{non-condensate modes}} \Rightarrow \langle a_r \rangle = \psi\, e^{i\bar{k}_0\bar{r}}$$

$$\Rightarrow \langle S^z_r \rangle = S - \bar{\psi}\, e^{-i\bar{k}_0\cdot\bar{r}}\, \psi\, e^{ik_0 r} = S - |\psi|^2 =$$

$$= S - \left(\frac{h_{sat} - h}{u}\right);$$

$$\langle S^+_r \rangle = \sqrt{2S}\, \psi\, e^{i\bar{k}_0\bar{r}} = \sqrt{2S}\, |\psi|\, e^{i\varphi + i\bar{k}_0\bar{r}}$$

$$\langle S^-_r \rangle = \sqrt{2S}\, \bar{\psi}\, e^{-i\bar{k}_0\bar{r}} = \sqrt{2S}\, |\psi|\, e^{-i\varphi + i\bar{k}_0\bar{r}}$$

$$\Rightarrow \langle S^x_r \rangle = \sqrt{2S}\, |\psi| \cos(\vec{k}_0\cdot\vec{r} + \varphi)$$

$$\langle S^y_r \rangle = \sqrt{2S}\, |\psi| \sin(\vec{k}_0\cdot\vec{r} + \varphi)$$

φ = phase of the order parameter

Note that $\vec{k}_0\cdot\vec{r} = \frac{\pi}{a}(1,1,1)\cdot(n_x a, n_y a, n_z a) =$

$$= \pi(n_x + n_y + n_z) = \begin{cases} \pi & \text{if } n_x + n_y + n_z = \text{odd} \\ 2\pi\cdot 0 & \text{if } \sum n = \text{even.} \end{cases}$$

Change of φ = rotation of $\langle S^{x,y} \rangle$ about z-axis $\Rightarrow$ does not cost energy

$\Rightarrow (\vec{r}, t)$ dependent fluctuations of φ represent low-energy modes.

Bose-Einstein condensation of magnons

$$H = \sum_{i,j} J_{ij} \vec{S}_i \cdot \vec{S}_j - g\mu_B B \sum_i S_i^z$$

Spin-to-boson mapping: $S^z = 1/2 - a^+a$, $S^+ = a$, $S^- = a+$

vacuum state $|O\rangle =$ for $h > h_{sat}(\{J_{ij}\})$

excited state = particle with momentum k : $|k\rangle = \frac{1}{\sqrt{N}} \sum_i S_i^- e^{ikr_i} |0\rangle = a_k^+ |0\rangle$

magnon = spin-flip = particle

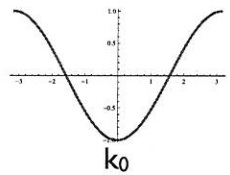
$$H = \sum_k (\varepsilon_k - \mu) a_k^+ a_k + \frac{1}{2} \sum_{ij} U_{ij} a_i^+ a_j^+ a_j a_i$$

chemical potential $\mu = h_{sat} - h$, condensation when $\mu = 0$

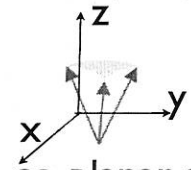
Magnons condense at k_0 (the minimum of spectrum): k_0 determines magnetic structure

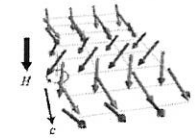
Magnon BEC

Single minimum k_0 = umbrella state = *superfluid*

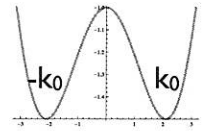


$$S_r^z = \frac{1}{2} - \rho \ , \ S_r^x = \sqrt{\rho}\cos[\vec{k}_0 \cdot \vec{r}] \ , \ S_r^y = \sqrt{\rho}\sin[\vec{k}_0 \cdot \vec{r}]$$

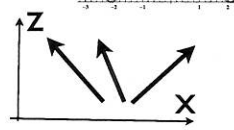


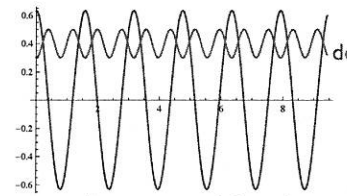


Non-equivalent k_0 and $-k_0$ = co-planar state = *supersolid*

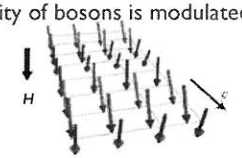


$$S_r^z = \frac{1}{2} - \rho - 2\rho\cos[2\vec{k}_0 \cdot \vec{r}] \ , \ S_r^x = \sqrt{\rho}\cos[\vec{k}_0 \cdot \vec{r}] \ , \ S_r^y = 0$$





density of bosons is modulated



There is a *quantum* phase transition between two types of condensates at some J'/J value